

EE 533P: Programming Practicum

Course Code	EE 533P
Course Name	Programming Practicum
L-T-P-C	1-0-3-3
Prerequisites	Programming experience in any language (C/C++/Matlab/Python etc.)
Intended for	MTech./M.S./Ph.D.
Distribution	Core for M.Tech. CSP, Elective for M.S./Ph.D.
Semester	Odd
Approval	recommended by the 30th BoA (Jun 2019)

Preamble

Programming skills are essential for any engineer. This course builds upon previous programming courses done by students during their previous degrees. A modern programming language with a large set of libraries suitable for data analysis or machine learning (such as Python) will be used. Emphasis will be given on programming for problems relevant to data analysis and information processing from domains such as signal processing, machine learning and communications.

Suggested implementation

The course will have weekly lab evaluations (related to the topic covered in the lecture), followed by a 3-4 week mini project. It is desirable for the the mini project to have a pre-final evaluation and a final evaluation.

Course Outcomes

After taking this course, students will become familiar in writing good code, using relevant libraries, using the right data structure, handling large data sets, plotting and visualizing information. They would have also become familiar with programming solutions to several problems in data analysis. They would also know how to do basic performance evaluation of programs.

Course Outline

1 lecture per week, followed by 3 hours of lab.

Course Modules

- **Review of general programming constructs:** Loops, conditionals, recursion, file i/o, data structures: strings, tuples, lists, dictionaries (4 lectures)
- **Introduction to scientific computing:** Numerical precision in programs, IEEE 754 floating point representation, introduction to NumPy and Scipy (1 lecture)
- **Data manipulation:** Pandas, handling large data files (1 lecture)
- **Data visualization:** Various types of plots: histograms, scatter plots, box plots etc. Datasets can be provided and plots can be created from them (1 lecture)
- **Object oriented programming:** Classes and objects, inheritance (1 lecture)
- **Implementing well known programs:** Matrix factorizations, solving large order linear systems of equations, least squares approximations, simulating binary channels, signal denoising, K-means clustering, classification using Baye's rule etc. (4 lectures)
- **Introduction to parallelization:** Cuda programming (conceptual level only) (1 lecture)
- **Program analysis:** Performance tuning, profiling of programs, identifying performance bottlenecks (1 lecture)

Text Books

1. *Python Programming for the Absolute Beginner*, Michael Dawson, Third Edition
2. *How to Think Like a Computer Scientist: Learning with Python*, Allen Downey, Jeffrey Elkner, Chris Meyers, Green Tea Press (Chapter 1-9, Chapter 11)

References

1. *Python Data Science Handbook*, Jake Vanderplas, O'Reilly
2. *Introduction to machine learning with Python*, Muller and Guido, O'Reilly